

Chapter 2: Setting up our Sandbox

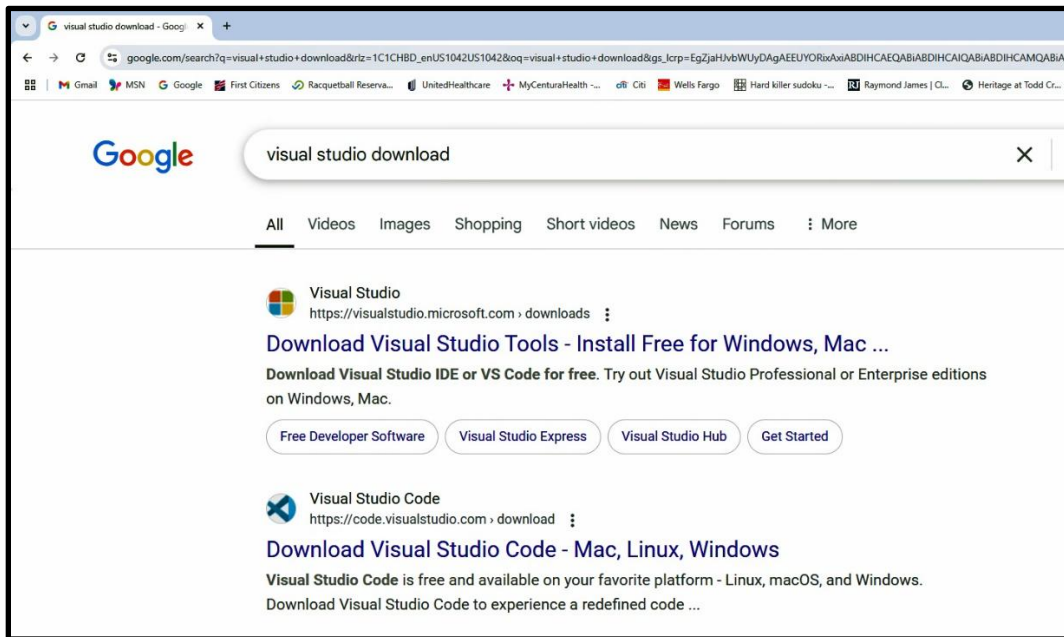
In this lesson we will be installing all the software tools used throughout the remainder of this course. Professional software developers often refer to that set of software development tools used daily as their sandbox, “hey, it’s time to go play in the sandbox.” Well, in this lesson we set up our sandbox. All these tools are free and very easy to install.

In this course we will be using 5 different pieces of software. Our main tool will be Microsoft’s Visual Studio. This tool is where we will design applications and write, debug and execute all our code. We will also be using Sequel Server and Sequel Server Management Studio from Microsoft. Sequel Server is a relational database and as the name implies Sequel Server Management Studio is a tool used to interface with that database. Relational databases are used for permanent data storage. We will also use XML Notepad from Microsoft. XML files are used as temporary data storage. We will use XML Notepad to learn how to read and write data to an XML file from within C#. Finally, we will use Oracle’s MySQL Workbench to create Entity-Relationship diagrams, graphic representations of a relational database design. As I mentioned, all these tools are free.

Here is the deal about installing software and teaching someone else how to do it. Both Microsoft and Oracle regularly change their install routines. I am going to give you instructions for how to best do these installs as of the summer of 2025. By the time you do these installations the exact methods may very well be different. I’m sorry, but please know you just need to get all five pieces of software installed. Stick with the vendor-supplied installation routines and you will be fine. I will give you the URL and screen shots for each starting point I used. Try to stick with those knowing that things can and will change in the software-install world over time.

Let’s begin by opening a browser and doing a search on the phrase “Visual Studio Download”. As shown here, several Microsoft download links come up. (Figure 2.1)

Figure 2.1: Visual Studio Download



I clicked on that first [visualstudio.microsoft.com > downloads](https://visualstudio.microsoft.com/downloads) link and got the page shown here. (Figure 2.2) From here you will want to click the button associated with the Community label. As soon as you open that page Microsoft downloads an application called VisualStudioSetup to your local computer and opens the page shown here. (Figure 2.3)

Figure 2.2: Visual Studio Download Page

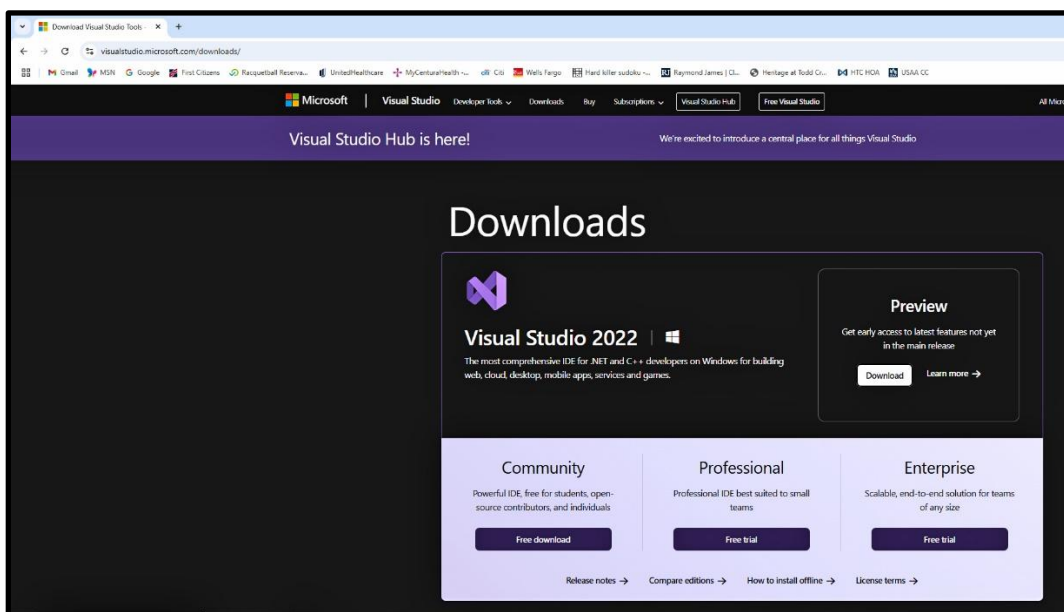
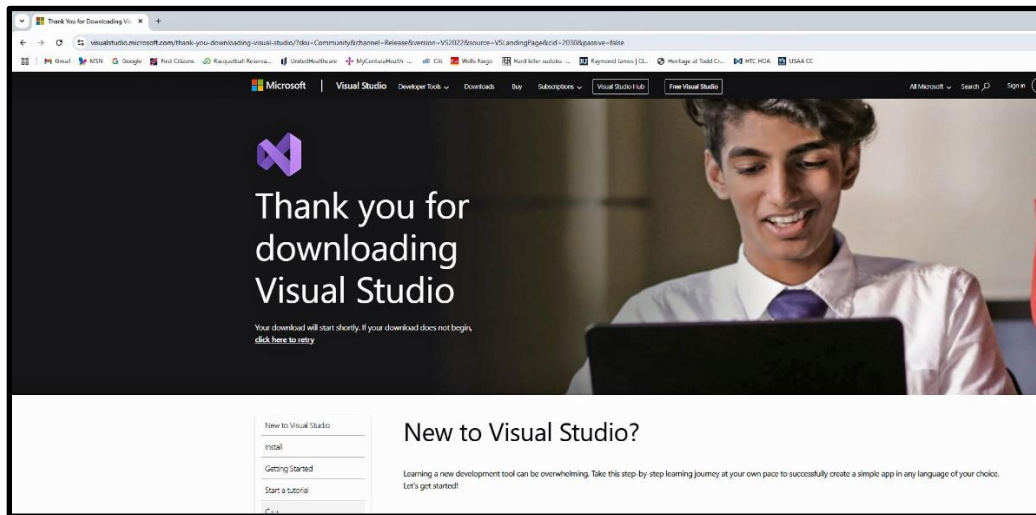
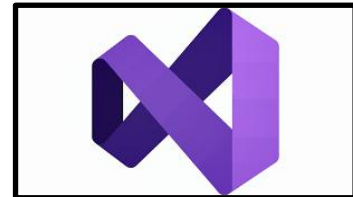


Figure 2.3: Visual Studio Install Page



Now this page clearly states that the installation of Visual Studio should start automatically at this point. Don't bet on it. It sure didn't on my box! Anyway, all you need to do is open File Explorer (or your operating system equivalent) and double click on the file named VisualStudioSetup in your Downloads folder. About 10-15 minutes later Visual Studio will be installed on your box and you will be looking at a new launch icon which looks like this purple stylized 3-D infinity symbol. The marketing people at Microsoft will tell you it represents a full and complete development cycle. I'm pretty sure they have no idea what that means.



We are going to use Visual Studio as our main tool for designing Forms and writing and debugging our C# code. In Chapter 3 we will open Visual Studio and design, code and debug our very first program, the always popular "Hello World" application.

Quick note here, if you didn't get a new launch icon just search on Visual Studio and open the application. When the application is open you should see the launch icon down in your taskbar. To pin it, just right click on it and select the appropriate option.

At this point Visual Studio should be ready to run so we can move onto MySQL Workbench. Guess what I Googled to find the download page for MySQL Workbench? Yes of course, MySQL Workbench Download. The associated image is shown here. (Figure 2.4) When I clicked the first dev.mysql.com > downloads > workbench link, I got the page shown here. (Figure 2.5) Please note the Oracle logo at the bottom of that page, if it is not there you are on an imposter's page. Never download from an imposter's webpage.

Figure 2.4: MySQL Workbench Download Search

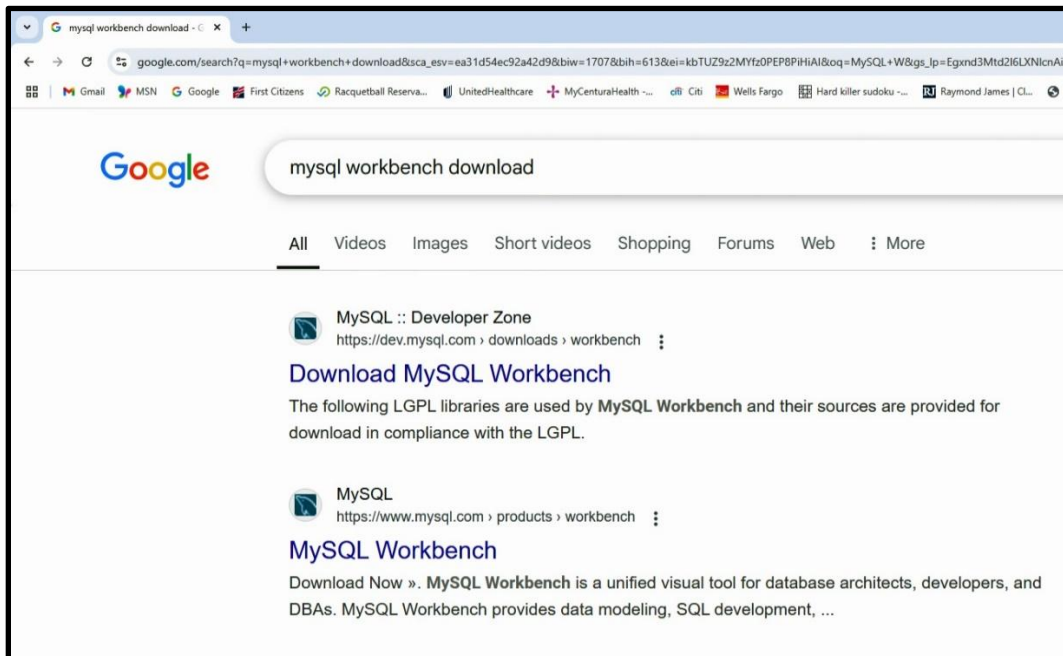
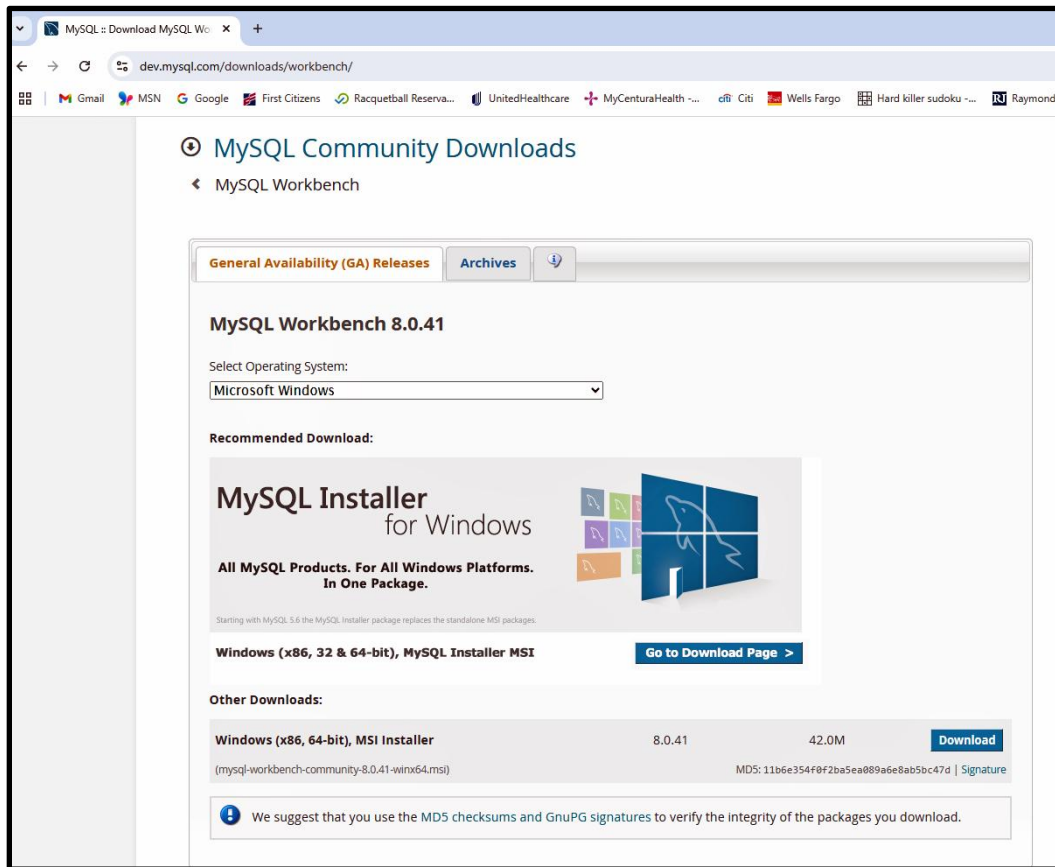


Figure 2.5: MySQL Workbench Download Page



When you click the download button Oracle will ask you to login into your Oracle Web Account. If you don't have an Oracle account, go ahead and create one. I know, it isn't cool to give out personal information on the web, but it is the only road to get to our goal so go ahead and give them your information.

When the download is complete you should find an installation routine in your download folder. Go ahead and double click that guy and the install will run. When the installation is complete MySQL Workbench will open and you can pin the launch icon to your taskbar.

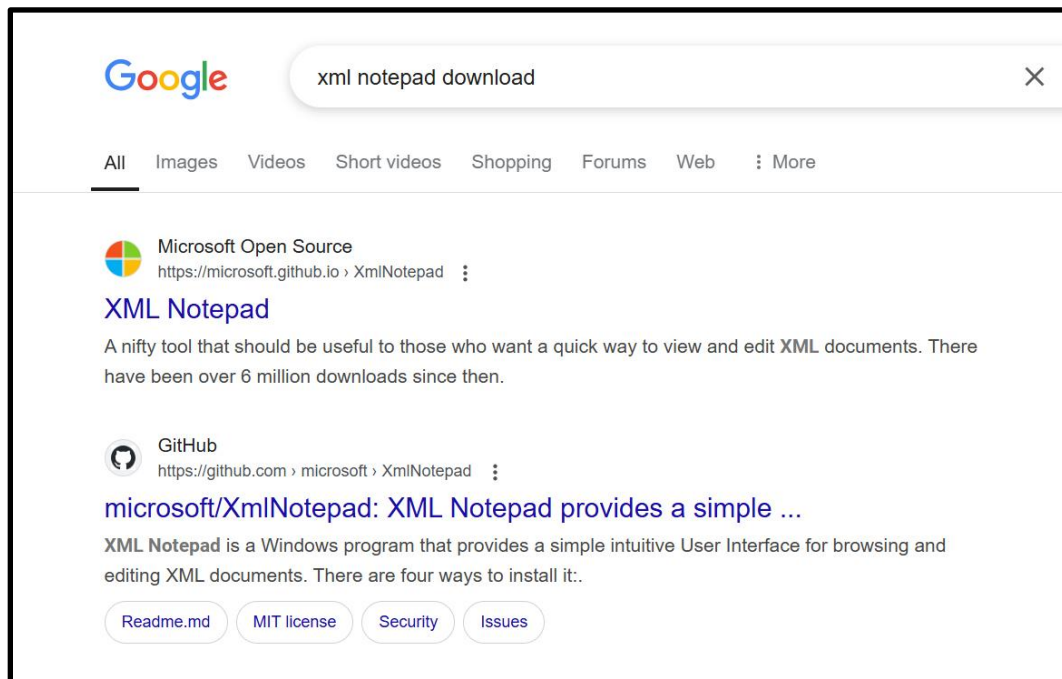
We will use MySQL Workbench to create Entity-Relationship diagrams when we study relational database fundamentals in Chapter 17.



We are next going to download XML Notepad from Microsoft. We will use this guy in Chapter 22 when we read and write data to an XML file. Unlike a relational database which is used for permanent storage of data, XML files are used as temporary storage mechanisms.

My search page is shown here. (Figure 2.6) To no one's surprise I simply did a search on XML Notepad Download and Microsoft responded appropriately. I chose the first link Microsoft.github.io > XmlNotepad.

Figure 2.6: XML Notepad Download Search Page



When I clicked that link, I got the page shown here. (Figure 2.7) From this page you are going to want to click the install link in the upper right-hand corner. When you do this, you will get the page shown here. (Figure 2.8)

Figure 2.7: XML Notepad Install Landing Page

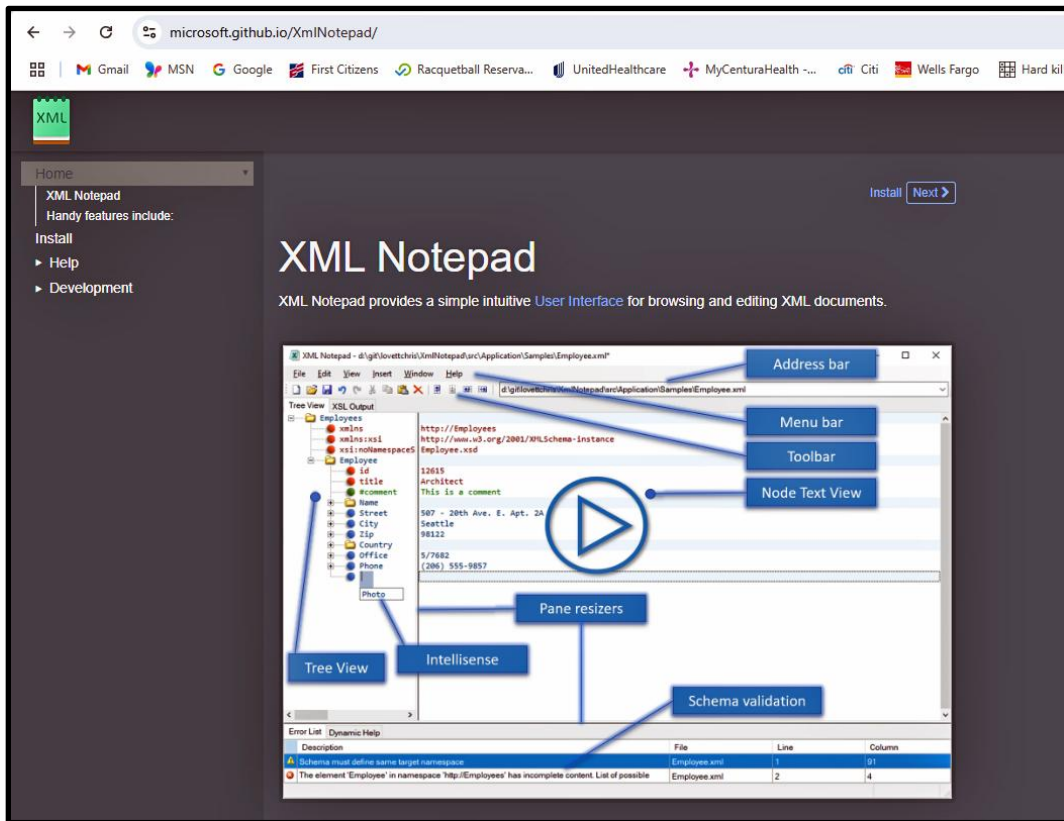
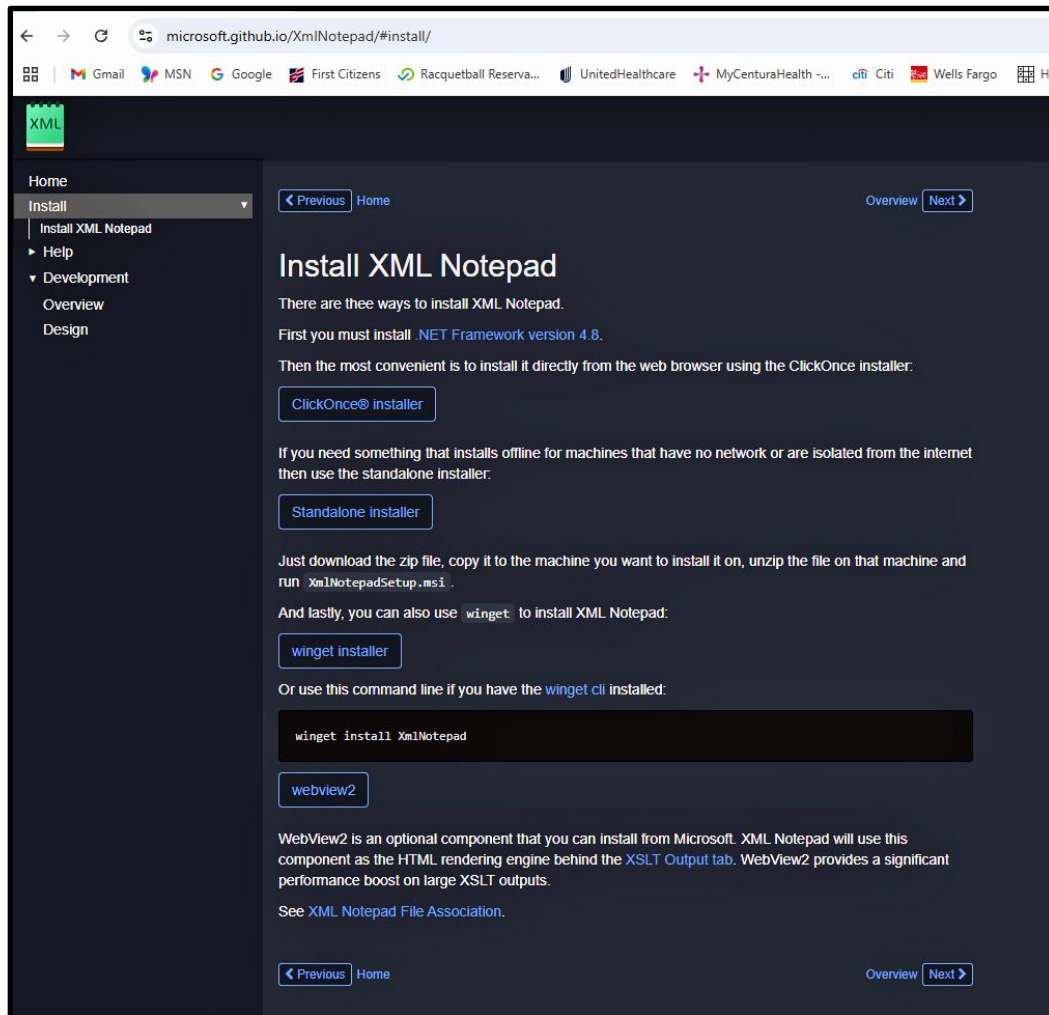


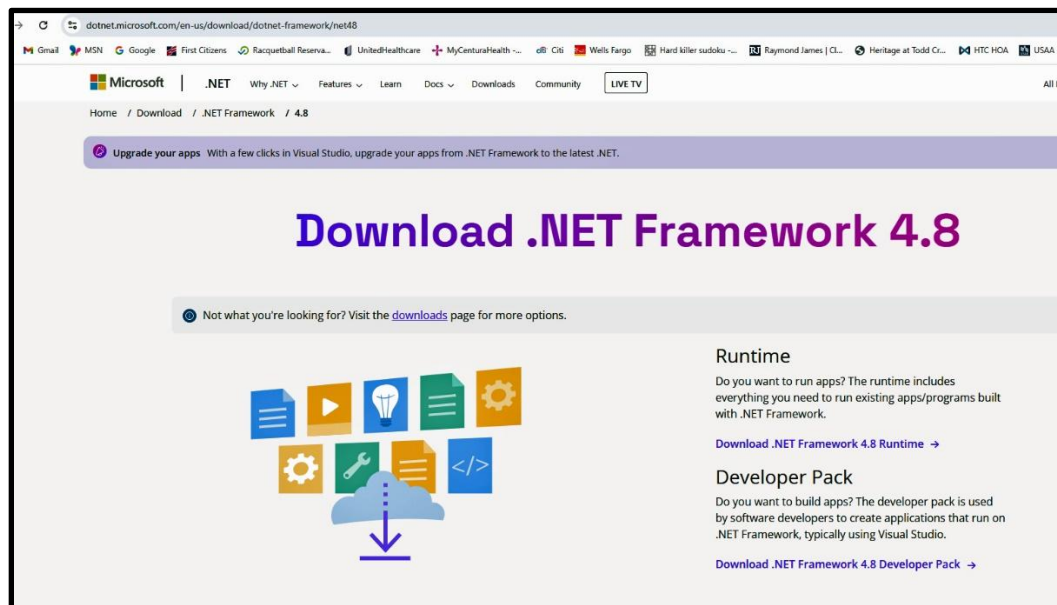
Figure 2.8: XML Notepad Install Page



Check it out, even mighty Microsoft makes mistakes. There is a typo in that first sentence. I am pretty sure that was supposed to read “There are three ways to install XML Notepad.” Not, “There are thee ways to install XML Notepad.” Well now I don’t feel so bad about not knowing how to use a semicolon in a sentence.

Anyway, as the page indicates, we first need to install the .NET Framework so let’s follow that link to the page shown here. (Figure 2.9) We just need to install the runtime version so go ahead and click that link to download the installer. The next step is to run that installer so please go to your downloads folder and double click on the file named ndp48-web. Again, please note that your download file may be named differently in the future so please be sure to check the download date to make certain you are running the correct installer. The .NET Framework installation will only take a couple of minutes after which you are ready to go back to the XML Notepad installation page shown previously. (Figure 2.8)

Figure 2.9: .Net Framework Install



Now from that installation page you can click the link for the ClickOnce Installer. (Figure 2.1) Now you should find a file named XMLNotepad in your downloads folder. That guy actually contains the application so just copy the whole thing to a location of your choice. Once you open the application you can pin the associated launch icon to your taskbar. The launch icon should look like this.



Our next piece of software is going to be Sequel Server. Sequel Server is Microsoft's relational database. We will use it in Chapters 18 through 21 when we learn how to read, write and update information within a relational database from within C#. Please note that we could have also used Oracle's Express Edition or MySQL (also from Oracle), but they are notorious for not installing properly so we will stick to Sequel Server which always seems to install okay.

My search page is shown here. (Figure 2.10) Please note that I specifically specified Microsoft in the search criterion as I know there are a ton of imposters out there for this software. The first link which came up was from the Trusted Tech Team which is supposed to be legit, but I went ahead and scrolled down to the link from Microsoft. The URL I followed was Microsoft.com > en-us > sql-server-downloads. Go ahead and click the title SQL Server Downloads to get to this page (Figure 2.11)

Figure 2.10: Sequel Server Download Search Results

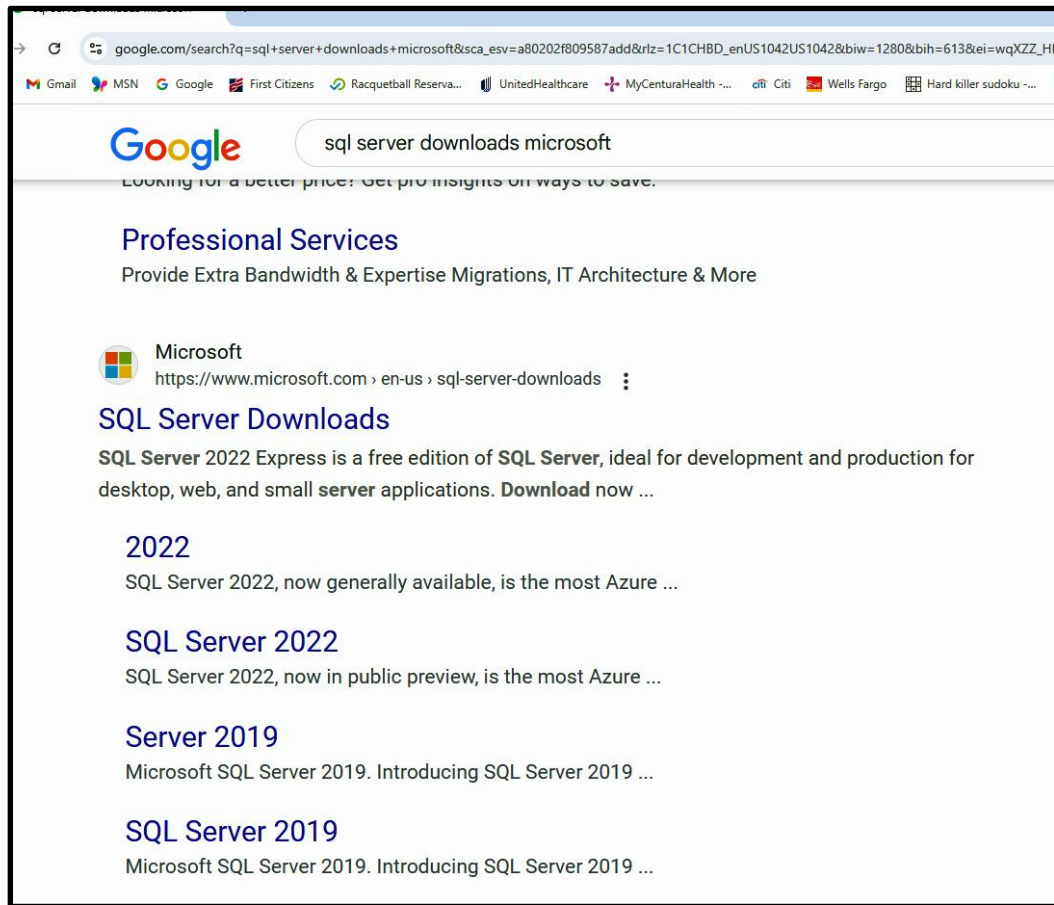
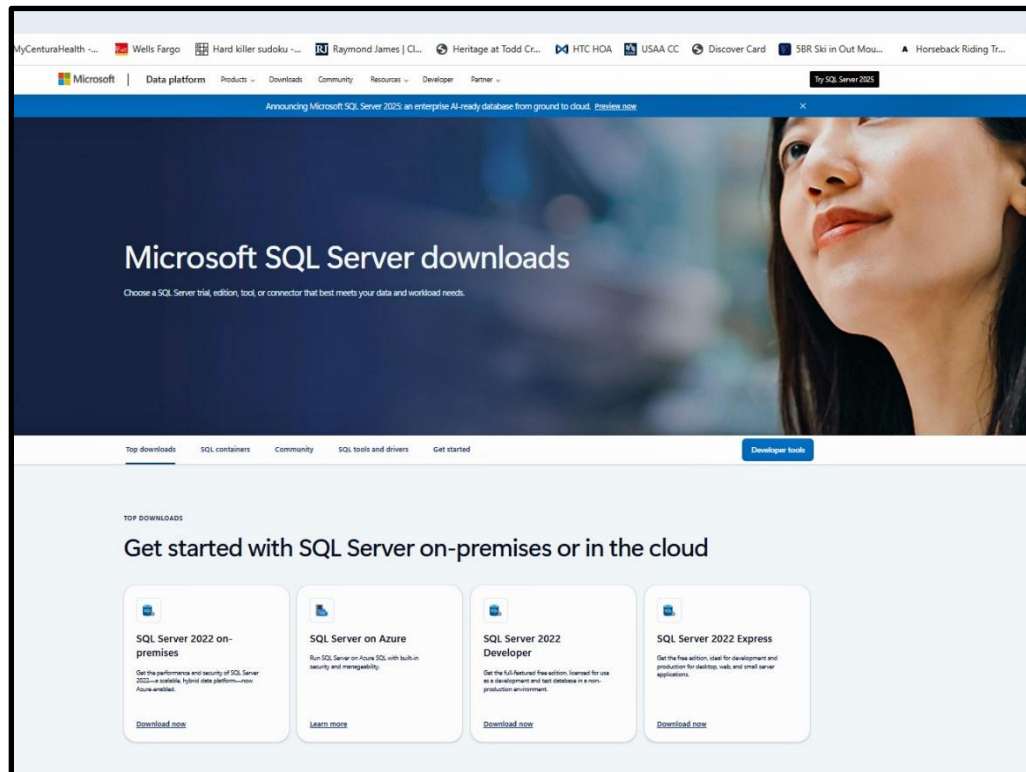
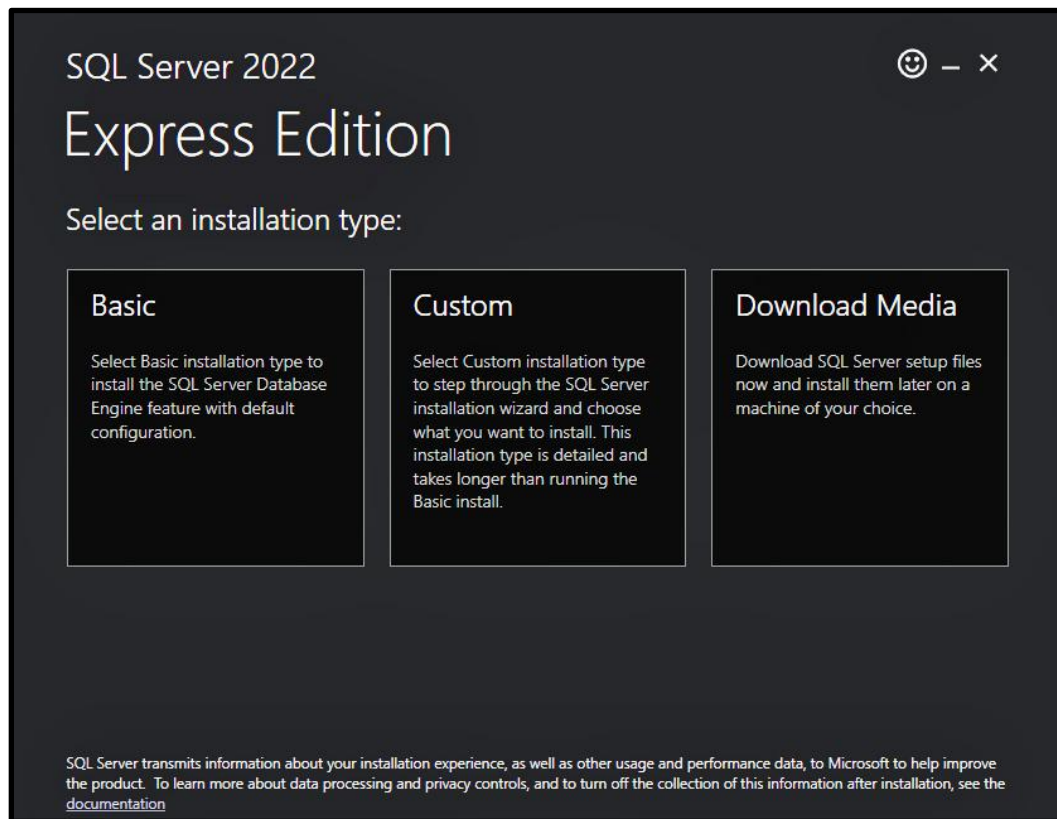


Figure 2.11: SQL Server Landing Page



The page shown here gives you all the available versions of Sequel Server. You should click on the Express edition download. Once you do so, you will find a file named SQL2022-SSEI-Expr in your downloads folder. Go ahead and double click that guy to bring up the install application. (Figure 2.12)

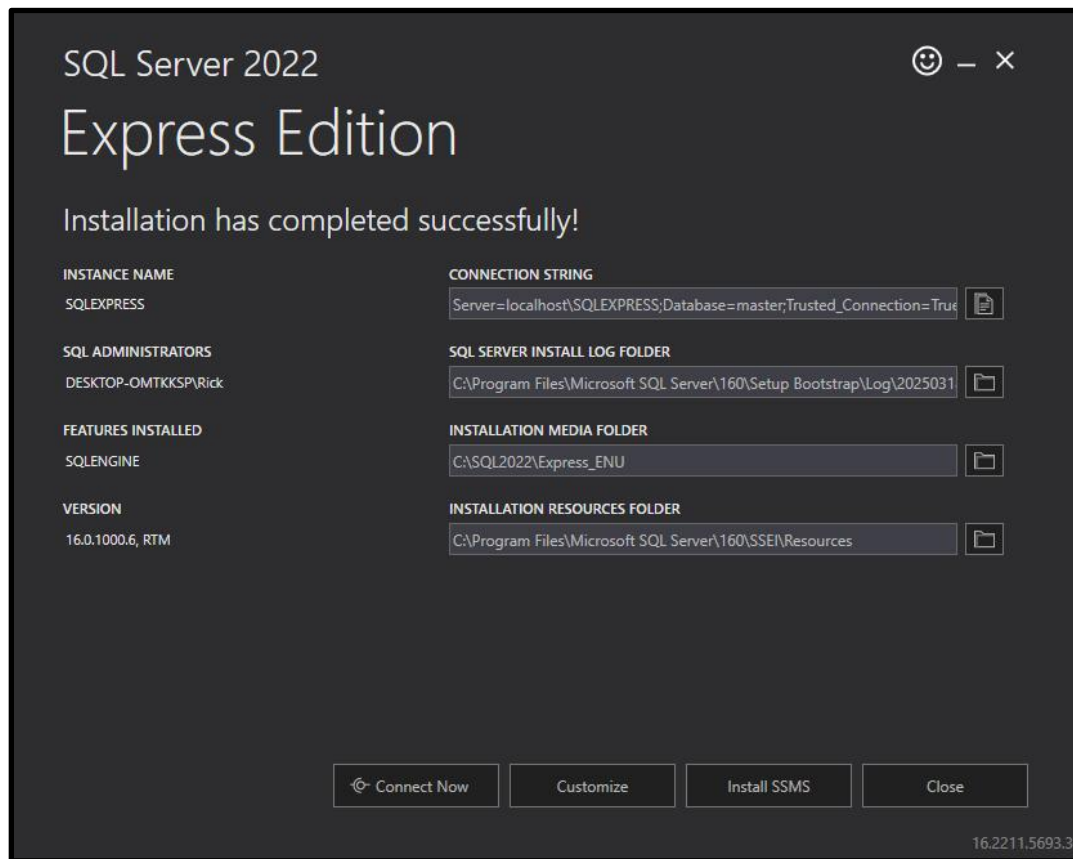
Figure 2.12: SQL Server Installer



On this screen, please click on the “Basic” option to start the install. Please note that there will not be a launch icon for Sequel Server. All relational databases run behind the scenes. We will use Sequel Server Management Studio, which we will install next, to manage our Sequel Server relational database.

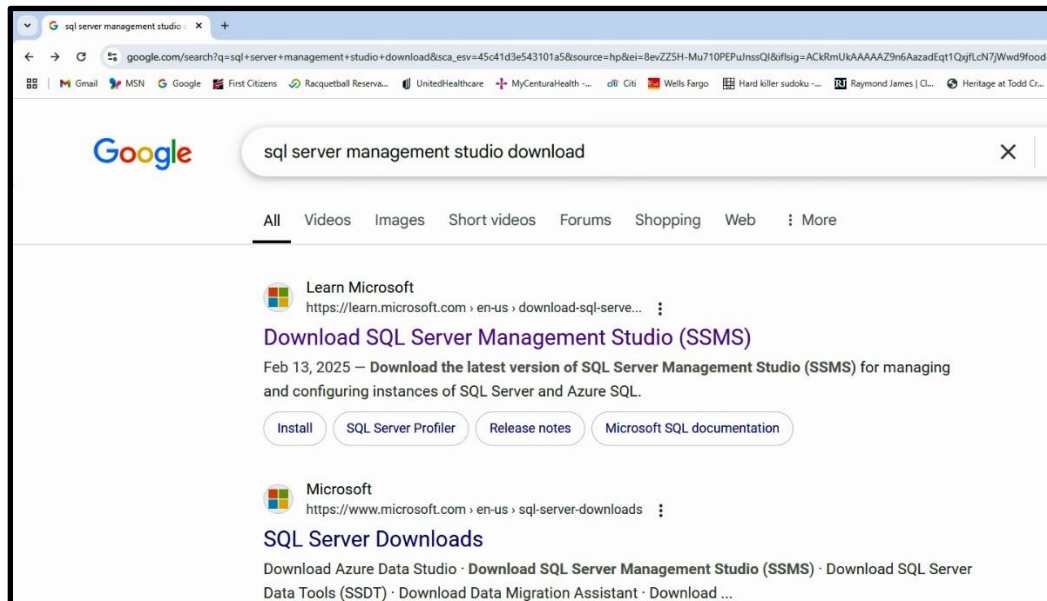
Once the installation is complete you will get the page shown here. (Figure 2.13) Always take a screenshot of this page and store it in a secure location because as you develop applications which interface with the database you are going to need to know that instance name, connection string and the administrator info.

Figure 2.13: Database Connection Information



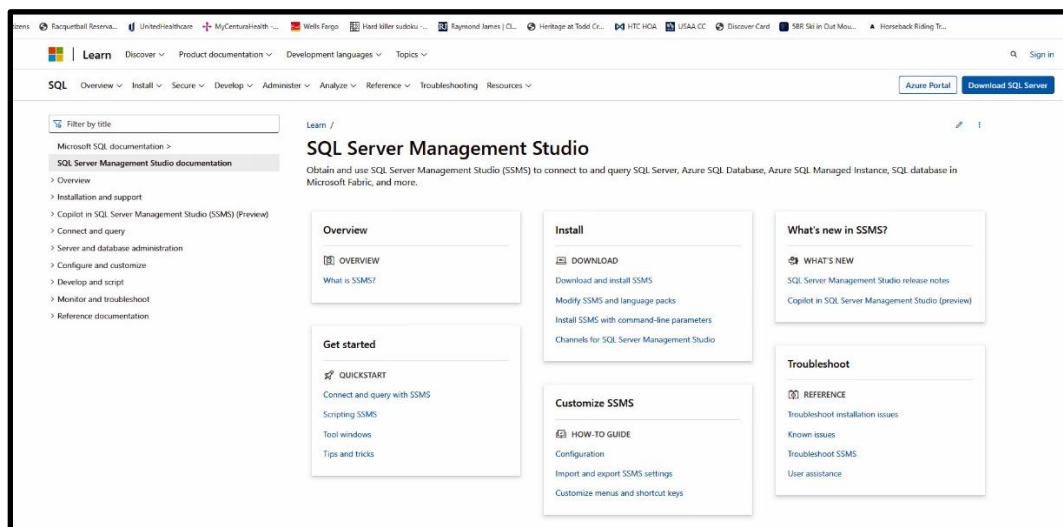
Our final piece of software will be Microsoft's Sequel Server Management Studio. We will use this software to execute stand-alone SQL commands in Chapter 18. As shown here my search was exactly as you would have expected. (Figure 2.14)

Figure 2.14: SQL Server Management Studio Download Search



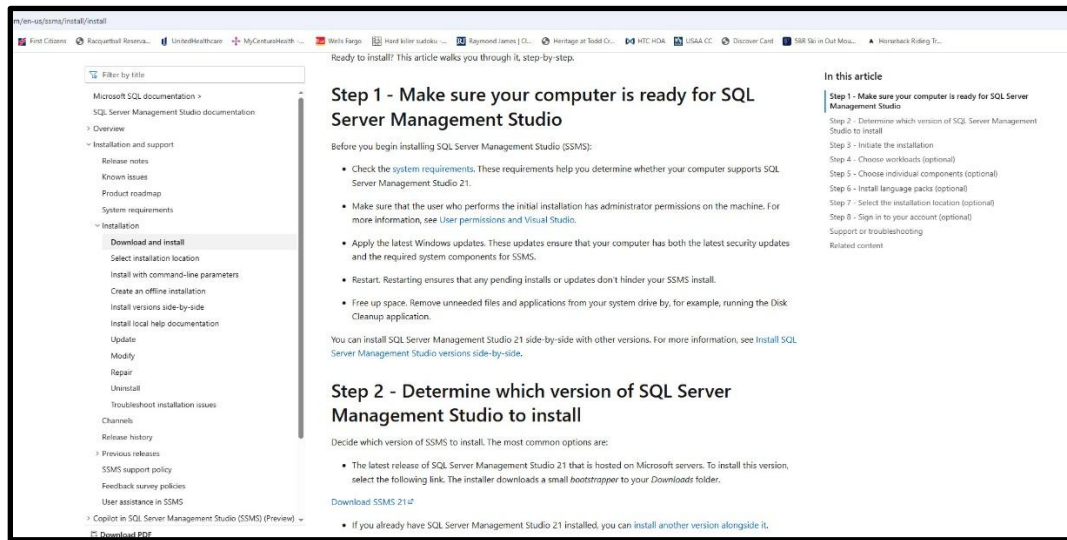
When I clicked the link labeled Install SQL Server Management Studio, I got the page shown here. (Figure 2.15)

Figure 2.15: SQL Server Management Studio Download Page



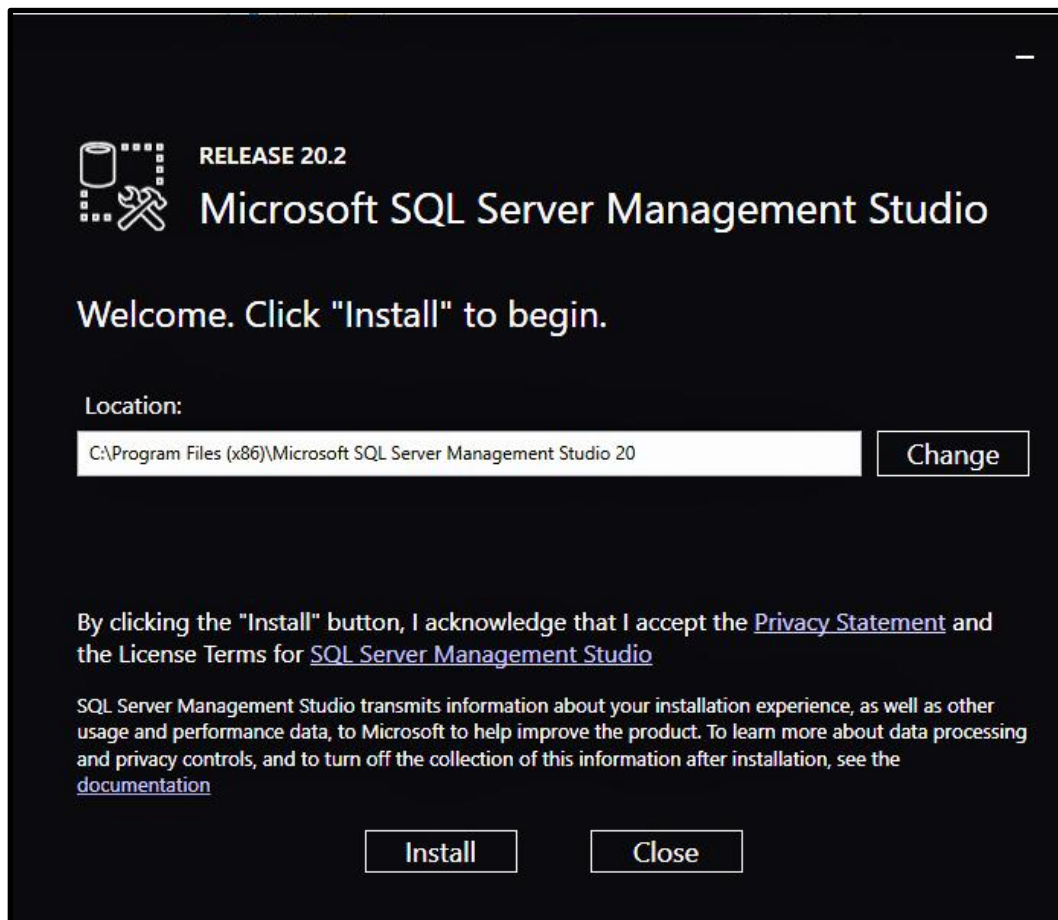
The next step is to click the “Download and install SSMS” link shown here. (Figure 2.16) At the time of this writing, I was downloading version 21, but you should download whatever version is current.

Figure 2.16: Download Link



Time to open your downloads folder and look for a file named SSMS-Setup-ENU. Double click that guy to start the installation. The page shown here comes up. (Figure 2.17) I just hit the “Install” button, but if you want to put the application somewhere different knock yourself out. Be aware, however, that the Management Studio software has to run on the same named drive as the database or the connection strings will not work.

Figure 2.17: SSMS Installation App



The launch icon for SSMS looks like the one shown here. Make sure to get a copy of that guy pinned to your taskbar.

Okay, this concludes our sandbox setup. At this point you should have Microsoft's Visual Studio, Microsoft's XML Notepad, Oracle's MySQL Workbench, and Microsoft's SQL Server and SQL Server Management Studio loaded on your computer. We are now ready to rock and roll in the next chapter where we will learn about Visual Studio and write and debug our first program. It should be great fun!



Again, I apologize for any install variations out there but please understand that it is not my doing. These guys simply don't work off the same page, and it ends up a big mess when you try to teach someone else how to do an installation. Anyway, I hope you didn't have too much trouble.